INEGOL INTEGRATED SOLID WASTE STORAGE AND DISPOSAL FACILITY

VCS Project Description - Template v4.4 draft

Generated from intake YAML and supplied methodology/template reference files.

Project title	INEGOL INTEGRATED SOLID WASTE STORAGE AND DISPOSAL FACILITY
Project ID	3908
Crediting period	31-December-2020 to 30-December-2027
Original date of issue	23-March-2023
Most recent date of issue	07-May-2026
Version	Draft 0.1
VCS Standard Version	4.7
Prepared by	Gaia Climate Finansal Danışmanlık Hizmetleri ve Ticaret A.Ş.

This draft uses project-specific facts from the intake file. Items not present in the source intake are marked as TBD and should be completed from the validated calculation spreadsheet, permits, stakeholder records, and supporting evidence package before submission.

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2 PROJECT DETAILS

2.1 Summary Description of the Project

The project activity is the INEGOL INTEGRATED SOLID WASTE STORAGE AND DISPOSAL FACILITY, an integrated municipal solid waste management and biogas-to-electricity project located in İnegöl District, Bursa Province, Türkiye. The source feedstock is described as: Municipal solid waste from Inegol, Iznik, and Yenişehir districts; organic fractions are sent to biomethanization; combustible fractions may be used for RDF production. Recyclable fractions are recovered where applicable, and residual fractions are managed through the waste management system.

The installed and operational electricity generation capacity is 8.484 MWe, consisting of six gas engines commissioned between 31-December-2020 and 03-November-2022. The generation license allows a total capacity of 14.140 MWe. Expected net electricity generation is 49,935.315 MWh per year and 349,547.209 MWh over the first crediting period.

The project reduces greenhouse gas emissions by diverting fresh municipal solid waste from disposal at a solid waste disposal site, avoiding methane emissions from anaerobic decay in the baseline, and generating renewable electricity from recovered biogas that is exported to the Turkish national grid via the İnegöl TM No-154/34.5 kV substation. The project applies ACM0022, version 03.0, for alternative waste treatment processes.

2.2 Audit History

Audit type	Period	Program	Validation/verification body name	Years
Validation	2020-12-31 to 2027-12-30	VCS	Earthood	7

2.3 Sectoral Scope and Project Type

Sectoral scope	01 Energy Industry - Renewable/Non-renewable Sources 13 Waste handling and disposal
Project activity type	Waste handling and disposal; non-AFOLU; not a grouped project.

2.4 Project Eligibility

2.4.1 General eligibility

The project is a non-AFOLU waste handling and disposal activity within the scope of the VCS Program. The intake identifies the project as listed on the Verra pipeline on 23-March-2023. The validation opening meeting date is recorded as 25-May-2023. Validation deadline exemptions are recorded as 24-November-2022, 23-June-2023.

The project activity is not identified as excluded under the VCS Standard in the intake. Final eligibility should be confirmed against the applicable VCS Standard version and the evidence package submitted to the VVB.

2.4.2 AFOLU project eligibility

Not applicable. The project is a non-AFOLU municipal solid waste management and energy generation project.

2.4.3 Transfer project eligibility

No transfer from another GHG program is identified in the intake. If any prior GHG program registration exists, the no-double-issuance evidence must be added before submission.

2.5 Project Design

The project is designed as a single location project. The intake identifies grouped project eligibility as not applicable.

2.5.1 Grouped project design

Not applicable. No grouped project instances or inclusion criteria are proposed in the source intake.

2.6 Project Proponent

Organization name	BIOTREND Çevre ve Enerji Yatırımları Anonim Şirketi
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Organization name	MUNDO VERDE CLIMATE SA
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Title	Director
Address	Rue Jean-Gabriel Eynard 8, 1205 Geneva, Switzerland
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Organization name	Gaia Climate Finansal Danışmanlık Hizmetleri ve Ticaret A.Ş.
Contact person	Gediz Kaya
Title	Managing Partner
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Email	gkaya@gaiaclimate.com

2.7 Other Entities Involved in the Project

Organization name	Doğu Star Elektrik Üretim A.Ş.
Role in the project	Facility Manager

Contact person	Aykut Mutlu
Title	Plant Manager
Address	Kavacık, Ekinciler Caddesi, Ertürk Sk. No:3, 34810 Beykoz/Istanbul
Telephone	+90 216 680 0000
Email	aykut.mutlu@biotrendenerji.com.tr

2.8 Ownership

The source PD states that, under the EMRA generation license dated 17-December-2020, all legal rights of the project belong to Doğu Star Elektrik Üretim A.Ş. It also states that MUNDO VERDE CLIMATE SA is the sole owner of carbon credits and that Gaia Climate is the authorized representative of MUNDO VERDE CLIMATE SA.

Ownership evidence identified in the intake: E001, E002, E003.

2.9 Project Start Date

Project start date	31-December-2020
Justification	The project start date is stated as 31-December-2020, corresponding to the partial acceptance date for the first gas engine and the commencement of electricity generation from biogas supplied to the national grid.

2.10 Project Crediting Period

Crediting period	Seven years, twice renewable
Start and end date of first crediting period	31-December-2020 to 30-December-2027

2.11 Project Scale and Estimated GHG Emission Reductions or Removals

The intake identifies the project scale class as <300000 tCO2e per year. The source project scale label is recorded as 100,001-1,000,000 tCO2e/year; this should be reconciled with the final emission reduction calculation spreadsheet before submission.

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO2e)
31-December-2020 to 31-December-2020	[TBD - source evidence/calculation required]
01-January-2021 to 31-December-2021	[TBD - source evidence/calculation required]
01-January-2022 to 31-December-2022	[TBD - source evidence/calculation required]
01-January-2023 to 31-December-2023	[TBD - source evidence/calculation required]
01-January-2024 to 31-December-2024	[TBD - source evidence/calculation required]
01-January-2025 to 31-December-2025	[TBD - source evidence/calculation required]

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO2e)
01-January-2026 to 31-December-2026	[TBD - source evidence/calculation required]
01-January-2027 to 30-December-2027	[TBD - source evidence/calculation required]
Total	[TBD - source evidence/calculation required]

2.12 Description of the Project Activity

The project includes a mechanical separation facility, RDF preparation facility, biomethanization facility, gas engines, booster unit, emergency flare, leachate handling, and connection to the Turkish national grid via Inegol TM No-154/34.5 kV substation. Six gas engines totaling 8.484 MWe are installed and operational; the license allows ten gas engines totaling 14.140 MWe.

The expected municipal solid waste throughput is 262,970.37 tonnes/year, equivalent to approximately 720.46 tonnes/day. The intake states that approximately 45.0% of the waste stream is suitable for biomethanization, corresponding to 324.2 tonnes/day.

The RDF preparation system has a maximum capacity of 27 tonnes/hour. Planned RDF throughput is stated as 93 tonnes/day in 2024 and 125 tonnes/day in 2035.

First gas engine partial acceptance occurred on 31-December-2020. Continuous on-site electricity generation started in June 2021. Gas engines were commissioned between 31-December-2020 and 03-November-2022. RDF fuel preparation facility commissioning is stated as 23-August-2023.

Engine	Model	Commissioning date
GM-1	INNIO JENBACHER JGS 420 GS-L	31-December-2020
GM-2	INNIO JENBACHER JGS 420 GS-L	01-October-2021
GM-3	INNIO JENBACHER JGS 420 GS-L	14-April-2022
GM-4	INNIO JENBACHER JGS 420 GS-L	01-July-2022
GM-5	INNIO JENBACHER JGS 420 GS-L	01-July-2022
GM-6	INNIO JENBACHER JGS 420 GS-L	03-November-2022

2.13 Project Location

The project site is located at Yeniyörük Mahallesi, Körkuyu; adjacent to the existing Solid Waste Landfill Facility of Bursa Metropolitan Municipality. The site area is 38,490.14 m2.

Country	Türkiye
Region/state	Bursa Province
City/district	İnegöl District
Grid connection point	İnegöl TM No-154/34.5 kV substation
Boundary evidence	E004

Point	Latitude	Longitude
1	40.1509707	29.5814230
2	40.1506186	29.5816029
3	40.1503405	29.5807083
4	40.1507017	29.5805288
5	40.1503495	29.5807087
6	40.1506234	29.5802686
7	40.1503230	29.5811724
8	40.1505138	29.5818071
9	40.1498528	29.5821439
10	40.1496622	29.5815105

2.14 Conditions Prior to Project Initiation

Prior to the project activity, fresh municipal waste was disposed of at a solid waste disposal site with partial landfill gas capture and flaring. The source PD states that the baseline scenario for fresh waste is disposal in a SWDS with partial LFG capture and flaring, and electricity generation in new grid-connected electricity plants.

The baseline service levels include disposal of municipal solid waste in the existing waste disposal system and electricity supplied by grid-connected electricity generation plants. The project activity provides alternative waste treatment, biogas recovery and use, and renewable electricity generation.

2.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The intake records an EMRA generation license dated 17-December-2020 and identifies Doğu Star Elektrik Üretim A.Ş. as holding the legal rights of the project. Final submission should include the generation license, environmental permits, waste management permits/municipal authorizations, grid connection documentation, and any EIA or EIA-exemption evidence applicable to the facility.

2.16 Double Counting and Participation under Other GHG Programs

2.16.1 No Double Issuance

No other GHG program registration is identified in the intake. Evidence to support no double issuance remains [TBD - source evidence/calculation required].

2.16.2 Registration in Other GHG Programs

No registration under another GHG program is identified in the intake.

2.16.3 Projects Rejected by Other GHG Programs

No rejection by another GHG program is identified in the intake.

2.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

2.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

No emissions trading program or binding emission limit interaction is identified in the intake. Final evidence is [TBD - source evidence/calculation required].

2.17.2 No Double Claiming with Other Forms of Environmental Credit

No other environmental crediting for the same GHG benefit is identified in the intake. Final evidence is [TBD - source evidence/calculation required].

2.17.3 Supply Chain (Scope 3) Emissions

The project reduces methane emissions from municipal solid waste management and displaces grid electricity. Any claims by downstream users of RDF, recycled materials, or electricity attributes should be reviewed and documented to prevent double claiming.

2.18 Sustainable Development Contributions

The project contributes to sustainable development by improving municipal waste handling, recovering recyclable and combustible fractions, treating organic waste through biomethanization, recovering biogas for renewable electricity generation, reducing methane emissions, and supporting local operational employment.

Sustainable development area	Project contribution	Monitoring approach
Climate mitigation	Avoided methane from fresh municipal waste and displacement of grid electricity.	Annual emission reduction calculations and electricity export/import records.
Clean energy	Renewable electricity from recovered biogas is exported to the Turkish national grid.	Metered electricity supplied to and consumed from the grid.
Responsible waste management	Mechanical separation, biomethanization, RDF preparation, and residual waste management improve integrated municipal waste treatment.	Waste reception records, process records, RDF production records, and residual disposal records.
Local employment and skills	The facility requires operational, maintenance, monitoring, and health and safety staff.	Employment and training records, where available.

2.19 Additional Information Relevant to the Project

2.19.1 Leakage Management

Leakage management is discussed in the source PD. Biogas goes directly to gas engines; flare equipment can destroy excess gas in emergency situations; pressure sensors detect unusual pressure drops; bypass lines, HDPE pipes, gas balloon storage, and safety valves are described as leak-management measures.

2.19.2 Commercially Sensitive Information

No commercially sensitive information is intentionally excluded from this draft except where source evidence is marked as TBD. Any final exclusions should be listed in Appendix 1 with justification.

2.19.3 Further Information

The final project description should be cross-checked against the validated calculation spreadsheet, permits, stakeholder consultation records, grid connection documents, equipment specifications, and ownership documentation.

3 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

3.1 Stakeholder Engagement and Consultation

3.1.1 Stakeholder Identification

Stakeholder Identification	Stakeholders expected to be relevant include Bursa Metropolitan Municipality and district municipalities, nearby residents and local administrative units around Yeniyörük Mahallesi, Körkuyu; adjacent to the existing Solid Waste Landfill Facility of Bursa Metropolitan Municipality., facility workers and contractors, waste collection/transport entities, grid/operator interfaces, recyclable/RDF off-takers, and relevant public authorities. Site-specific stakeholder lists and evidence are TBD.
Legal or customary tenure/access rights	The project is located adjacent to the existing solid waste landfill facility on a defined site area of 38,490.14 m2. Evidence of land rights, municipal access rights, and any customary access considerations is [TBD - source evidence/calculation required].
Stakeholder diversity and changes over time	Stakeholders may include public authorities, local communities, facility workers, waste sector contractors, and commercial off-takers. Demographic and vulnerability information is TBD.
Expected changes in well-being	Expected effects include improved waste management, reduced landfill methane potential, employment, and possible localized traffic, odour, noise, and occupational health and safety risks that require management.
Location of stakeholders	Primary stakeholders are expected to be located in and around the project site and the service area described in the intake: Municipal solid waste from Inegol, Iznik, and Yenişehir districts; organic fractions are sent to biomethanization; combustible fractions may be used for RDF production.
Location of resources	Relevant resources include the project site, municipal waste streams, access roads, grid connection infrastructure, and surrounding community resources. Site-specific mapping evidence is TBD.

3.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	[TBD - source evidence/calculation required]
Stakeholder engagement process	Stakeholder consultation records were not included in the intake. The final PD should describe meeting announcements, participant groups, language/accessibility arrangements, meeting minutes, attendance records, and how input was documented.
Consultation outcome	[TBD - source evidence/calculation required]

Ongoing communication	The final PD should provide contact points, grievance channels, response timelines, and records management procedures for ongoing communication with stakeholders.
Stakeholder input	No site-specific input was provided in the intake. Final design updates or justification for no updates should be added after reviewing consultation evidence.

3.1.3 Free Prior and Informed Consent

Obtaining consent	No Indigenous Peoples, local communities with customary tenure rights, or FPIC-triggering circumstances are identified in the intake. This conclusion should be confirmed with stakeholder mapping and land/access rights evidence.
Outcome of FPIC	Not applicable based on the intake, subject to confirmation through stakeholder identification and land/access rights review.

3.1.4 Grievance Redress Procedure

Development process	[TBD - source evidence/calculation required]
Grievance redress procedure	The final PD should include the procedure for receiving, recording, investigating, responding to, and closing grievances, including responsible persons, response timelines, appeal/escalation process, and records retention.

3.1.5 Public Comments

Comments received	Actions taken
[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]

3.2 Risks to Stakeholders and the Environment

3.2.1 Management Experience

The project proponent group and facility manager are identified in the intake. Final submission should include evidence of experience in waste management, power generation, occupational health and safety, environmental compliance, and VCS project implementation.

3.2.2 Risk Assessment

Risk category	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Odour, traffic, noise, fire/explosion risk from biogas, and emergency events.	Operate gas collection, booster, flare, safety valves, pressure sensors, emergency procedures, traffic controls, and community communication channels.

Risk category	Risks identified	Mitigation or preventative measure(s) taken
Risks to stakeholder participation	Low access to information or limited awareness of consultation channels.	Provide clear public contact points, accessible consultation notices, and grievance channels.
Environmental impacts	Leachate, wastewater, residual waste, air emissions, and odour.	Operate leachate handling, process controls, permitted discharge/treatment systems, regular inspections, and environmental monitoring.
Occupational health and safety	Mechanical equipment, confined spaces, biogas, electrical equipment, and vehicle movements.	Use training, PPE, lockout/tagout, gas detection, permit-to-work procedures, and emergency drills.
Project continuity	Equipment downtime and flare/engine interruptions.	Preventive maintenance, emergency flaring, spare parts management, and monitoring of engine and gas system performance.

3.3 Respect for Human Rights and Equity

3.3.1 Labor and Work

Risk category	Risks identified	Mitigation or preventative measure(s) taken
Discrimination	No specific risk identified in the intake.	Apply equal opportunity employment practices and grievance channels.
Sexual harassment	No specific risk identified in the intake.	Apply workplace conduct policies, confidential reporting, and disciplinary procedures.
Forced labor	No specific risk identified in the intake.	Comply with Turkish labor law and supplier/contractor requirements.
Child labor	No specific risk identified in the intake.	Verify worker age and contractor compliance.
Freedom of association	No specific risk identified in the intake.	Comply with applicable labor rights and maintain worker communication channels.
Occupational health and safety	Biogas, mechanical, traffic, and electrical hazards.	Training, PPE, inspections, emergency response, and permit-to-work controls.

3.3.2 Human Rights

No human rights risk specific to the project is identified in the intake. Final submission should document the screening process and applicable mitigation measures.

3.3.3 Indigenous Peoples and Cultural Heritage

No Indigenous Peoples or cultural heritage impacts are identified in the intake. This should be confirmed with stakeholder mapping and permitting evidence.

3.3.4 Property Rights

No displacement or adverse property-right impact is identified in the intake. Land tenure, access rights, and municipal authorization evidence should be attached or referenced.

3.3.5 Benefit Sharing

No separate benefit-sharing plan is identified in the intake. The project benefits are expected to arise through improved municipal waste management, employment, renewable electricity generation, and climate mitigation. If affected stakeholder groups require a benefit-sharing plan, it should be added.

3.4 Ecosystem Health

The project is a waste management and energy project at/adjacent to an existing solid waste facility, not an AFOLU land-use project. The final PD should confirm site ecological screening, permitting, and any required environmental management measures.

Risk category	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No site-specific biodiversity risk identified in the intake.	Confirm through permit/EIA screening and operate within permitted project footprint.
Soil degradation and soil erosion	Potential localized risk during construction/maintenance.	Use erosion controls, paved/managed operational areas, and stormwater/leachate management.
Water consumption and stress	Water use and wastewater/leachate handling require control.	Monitor water use, collect/treat leachate and wastewater according to permits.
Pollution	Potential odour, air emissions, leachate, and noise.	Apply process controls, biogas management, emergency flare, and environmental monitoring.

3.4.1 Rare, Threatened, and Endangered Species

No rare, threatened, or endangered species information was provided in the intake. Site-specific screening is [TBD - source evidence/calculation required].

3.4.2 Introduction of Species

Not applicable. The project does not involve planting, species introduction, or monoculture establishment.

3.4.3 Ecosystem Conversion

Not applicable as an AFOLU conversion requirement; nevertheless, final permitting evidence should confirm that the project footprint is authorized and does not convert protected natural ecosystems.

4 APPLICATION OF METHODOLOGY

4.1 Title and Reference of Methodology

Type	Reference ID	Title	Version
Methodology	ACM0022	Alternative waste treatment processes	03.0
Tool	TOOL02	Combined tool to identify the baseline scenario and demonstrate additionality	07.0
Tool	TOOL03	Tool to calculate project or leakage CO2 emissions from fossil fuel combustion	03.0
Tool	TOOL04	Emissions from solid waste disposal sites	08.1
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL08	Tool to determine the mass flow of a greenhouse gas in a gaseous stream	03.0
Tool	TOOL14	Project and leakage emissions from anaerobic digesters	02.0
Tool	TOOL24	Common practice	03.1
Tool	TOOL27	Investment analysis	14.0

4.2 Applicability of Methodology

Methodology/tool	Applicability condition	Justification of compliance
ACM0022	Project installs and operates new plant(s) for treatment of fresh waste through eligible processes such as anaerobic digestion with biogas recovery and/or mechanical/thermal treatment to produce RDF/SB.	The project includes mechanical separation, biomethanization, biogas recovery/use in gas engines, emergency flaring, and RDF preparation for municipal solid waste.

Methodology/tool	Applicability condition	Justification of compliance
ACM0022	Fresh waste would otherwise be disposed in a solid waste disposal site, with or without partial LFG capture.	Prior to the project activity, fresh municipal waste was disposed of at a solid waste disposal site with partial landfill gas capture and flaring. The source PD states that the baseline scenario for fresh waste is disposal in a SWDS with partial LFG capture and flaring, and electricity generation in new grid-connected electricity plants.
ACM0022	Fresh waste and products from the project plant are not stored on-site under anaerobic conditions outside the eligible treatment process.	The intake describes direct biogas routing to engines, emergency flare availability, gas balloon storage, bypass lines, pressure sensors, and safety valves. Final operating procedures should confirm storage conditions.
ACM0022	Wastewater discharge from the project activity is treated according to applicable regulations.	Leachate handling is included in the facility description. Permit and discharge evidence is [TBD - source evidence/calculation required].
ACM0022	The project activity does not reduce recycling that would occur in the absence of the project.	The mechanical separation facility recovers recyclable fractions where applicable. Baseline recycling evidence is [TBD - source evidence/calculation required].
ACM0022	Hazardous wastes/wastewater are not eligible.	The feedstock is municipal solid waste from the named districts. Final waste acceptance procedures should confirm exclusion of hazardous waste.
TOOL02	Baseline scenario identification and additionality demonstration are required.	The project uses the combined tool; final baseline selection and additionality evidence are to be completed with investment/common practice analysis.
TOOL14	Anaerobic digester project and leakage emissions are addressed where anaerobic digestion is used.	The project uses biomethanization and biogas recovery; monitored biogas and operational data will support calculations.
TOOL05/TOOL07	Electricity consumption/generation and grid emission factor are addressed where electricity is consumed or exported.	The project exports renewable electricity to the Turkish grid and consumes auxiliary electricity; metering data and applicable grid EF are required.

4.3 Project Boundary

The project boundary includes the solid waste disposal site baseline for waste that would otherwise be disposed, the project alternative waste treatment facilities, on-site biogas management and electricity generation equipment, on-site fuel and electricity use, wastewater/leachate handling associated with the treatment process, and the Turkish grid plants connected to the electricity system displaced by project electricity exports. Waste collection and transport are excluded from the boundary under ACM0022.

Boundary element	Included project facilities/processes
Waste reception and mechanical separation	Municipal solid waste from Inegol, Iznik, and Yenişehir districts; organic fractions are sent to biomethanization; combustible fractions may be used for RDF production.; sorting/separation of organic, recyclable, combustible, and residual fractions.

Boundary element	Included project facilities/processes
Biomethanization	Anaerobic treatment of organic fractions, biogas recovery, gas handling, storage/safety devices, and associated digestate/leachate handling.
Electricity generation	Six gas engines totaling 8.484 MWe, booster unit, metering, and connection to the Turkish national grid.
Emergency flaring	Emergency flare for destruction of excess biogas when engines cannot consume all produced gas.
RDF preparation	Mechanical/thermal preparation of combustible fractions for RDF, with residual waste management.
Excluded	Waste collection and transport outside the facility boundary, unless required by a specific leakage calculation.

Scenario	Source	Gas	Included?	Justification
Baseline	SWDS disposal of fresh waste	CH_4	Yes	Major methane source avoided by alternative waste treatment.
Baseline	SWDS disposal of fresh waste	CO_2	No	Biogenic CO_2 from decomposition of fresh waste is not accounted under ACM0022.
Baseline	SWDS disposal of fresh waste	N_2O	No	Excluded as small compared with CH_4 and conservative.
Baseline	Electricity generation displaced by project exports	CO_2	Yes	Project electricity exported to the grid displaces grid-connected generation.
Baseline	Electricity generation	CH_4/N_2O	No	Excluded for simplification/conservativeness under the methodology.
Project	On-site fossil fuel consumption	CO_2	Yes	Included where fossil fuel is consumed by project equipment.
Project	On-site electricity use	CO_2	Yes	Auxiliary electricity consumption may be an important project emission source.
Project	Anaerobic digestion and gas handling	CH_4	Yes	Potential methane leakage from digesters/gas systems and incomplete combustion.
Project	Flaring	CH_4	Yes	Incomplete combustion from emergency flaring is addressed using the relevant tool.

Scenario	Source	Gas	Included?	Justification
Project	RDF/SB production and use	CO2/CH4/N2O	Yes/No as applicable	Included where fossil-based fractions or off-site end use create emissions required by ACM0022; final treatment and end-use evidence is required.
Leakage	Anaerobic digestion and RDF/SB by-products/end use	CH4/CO2/N2O	Yes as applicable	Leakage sources are treated according to ACM0022 and relevant tools.

4.4 Baseline Scenario

The baseline scenario is selected using ACM0022 and the combined tool to identify the baseline scenario and demonstrate additionality. Based on the intake, the baseline for fresh municipal solid waste is disposal in a SWDS with partial landfill gas capture and flaring. For electricity generated by the project activity, the baseline is electricity generation in existing and/or new grid-connected electricity plants.

- *Waste baseline: fresh municipal solid waste disposed at a SWDS with partial LFG capture and flaring.*
- *Energy baseline: electricity supplied by grid-connected electricity plants serving the Turkish national grid.*
- *Project alternative: alternative treatment of municipal solid waste through mechanical separation, biomethanization with biogas recovery and use, RDF preparation, and renewable electricity generation.*

4.5 Additionality

4.5.1 Regulatory Surplus

The intake does not identify a regulation mandating the project activity at a compliance rate that would make the activity non-additional. Final regulatory surplus evidence should include applicable Turkish waste, renewable energy, landfill gas, environmental, and licensing requirements and demonstrate that the project activity is not legally required or is eligible under ACM0022 where compliance rates are below the applicable threshold.

4.5.2 Additionality Methods

Additionality is to be demonstrated using the combined tool to identify the baseline scenario and demonstrate additionality, supported by the investment analysis tool and common practice tool listed in the intake. The final PD should include the selected analysis route, all inputs and sources, sensitivity analysis, and common practice assessment sufficient for a reader to reproduce the conclusion.

Step	Draft application	Evidence status
Identify realistic and credible alternatives	Alternatives include continued disposal in the SWDS with partial LFG capture/flaring, implementation of alternative waste treatment without carbon revenue, and project implementation with VCS revenue.	[TBD - source evidence/calculation required]
Investment analysis	Apply TOOL27 to assess financial attractiveness of the project activity without carbon revenue.	[TBD - source evidence/calculation required]
Barrier analysis, if selected	Not selected in this draft; to be confirmed.	[TBD - source evidence/calculation required]

Step	Draft application	Evidence status
Common practice	Apply TOOL24 to assess similar integrated MSW/biogas/RDF projects in Türkiye.	[TBD - source evidence/calculation required]
Conclusion	The project is expected to be additional subject to completion of investment and common practice evidence.	[TBD - source evidence/calculation required]

4.6 Methodology Deviations

No methodology deviations are identified in the intake. If any deviation is applied during validation, it must be described and justified here, including evidence that it does not reduce conservativeness and relates only to monitoring or measurement criteria/procedures.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions are quantified in accordance with ACM0022. For this project, relevant baseline components include methane emissions from fresh waste that would have been disposed in the baseline SWDS and baseline emissions from electricity generation displaced by renewable electricity exported to the grid.

- *BE_y = sum of applicable baseline components for each treatment process and year, adjusted for any applicable regulatory compliance discount factor.*
- *BE_CH4,SWDS,y is calculated using TOOL04 for the amount and composition of fresh waste prevented from disposal.*
- *BE_EC,y is calculated using the applicable electricity generation/consumption tool and Turkish grid emission factor for net electricity supplied by the project.*

5.2 Project Emissions

Project emissions include applicable emissions from anaerobic digestion, gas handling, flaring, on-site fossil fuel use, on-site electricity consumption, RDF/SB production and end-use treatment where required, and any other sources included by ACM0022 and the referenced tools.

- *PE_AD,y is determined using the tool for project and leakage emissions from anaerobic digesters.*
- *PE_flare,y is determined using the tool for project emissions from flaring where emergency flaring occurs.*
- *PE_EC,y accounts for auxiliary electricity consumption from the grid.*
- *PE_FC,y accounts for fossil fuel consumption by project equipment, if any.*

5.3 Leakage Emissions

Leakage emissions are assessed for anaerobic digestion and RDF/SB pathways according to ACM0022. The intake identifies leakage management measures including direct biogas routing to engines, emergency flaring, pressure sensors, bypass lines, HDPE pipes, gas balloon storage, and safety valves. Final leakage calculations require monitored or estimated data on digestate/by-products, RDF/SB end use, residual disposal, and any off-site anaerobic decomposition risks.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Emission reductions are calculated as baseline emissions minus project emissions and leakage emissions. Carbon dioxide removals are not claimed by this non-AFOLU waste management project.

The intake provides expected electricity generation but does not provide annual baseline, project, leakage, or net emission reduction values. The table below is therefore prepared with required vintage periods and marked [TBD - source evidence/calculation required] pending the calculation spreadsheet.

Vintage period	Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated reductions (tCO2e)	Estimated removals (tCO2e)	Estimated VCUs (tCO2e)
31-Dec-2020 to 31-Dec-2020	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2021 to 31-Dec-2021	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]

Vintage period	Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated reductions (tCO2e)	Estimated removals (tCO2e)	Estimated VCUs (tCO2e)
01-Jan-2022 to 31-Dec-2022	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2023 to 31-Dec-2023	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2024 to 31-Dec-2024	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2025 to 31-Dec-2025	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2026 to 31-Dec-2026	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
01-Jan-2027 to 30-Dec-2027	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]
Total	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	[TBD - source evidence/calculation required]	0	[TBD - source evidence/calculation required]

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	Unit	Description	Value applied	Source	Comments
Installed_capacity	MW	Installed operational electricity generation capacity	8.484	Project technical/equipment records in intake	Capacity should be verified against equipment commissioning records and license.
License_capacity	MW	Licensed total electricity generation capacity	14.140	EMRA generation license referenced in intake	License evidence to be attached.
Expected_waste_throughput	tonnes/year	Expected municipal solid waste throughput	262,970.37	Source PD/intake	Used for ex-ante estimates; monitored values supersede during verification.
Biomethanization_suitable_fraction	fraction	Expected fraction suitable for biomethanization	0.45	Source PD/intake	Waste characterization evidence required.
Project_start_date	date	Project start date	31-December-2020	Partial acceptance date for first gas engine	Used for crediting period and vintage boundaries.
Grid_connection_point	text	Grid connection point	İnegöl TM No-154/34.5 kV substation	Source PD/intake	Grid connection documentation required.

6.2 Data and Parameters Monitored

Data / Parameter	Unit	Description	Frequency	Monitoring equipment/source	QA/QC procedures
W_MSW,y	tonnes	Municipal solid waste received/treated by the project	Continuous/weighbridge or operational records; aggregated monthly and annually	Weighbridge/waste acceptance system	Calibration and reconciliation with municipal records
Waste fractions	fraction or %	Composition of waste by relevant ACM0022 categories	Sampling/ characterization according to approved procedure	Waste characterization records/lab analysis	Sampling plan and QA/QC required
EG_export,y	MWh	Electricity supplied to the grid	Continuous meter reading; monthly and annual aggregation	Revenue/export meter	Meter calibration, cross-check with grid invoices
EG_import,y	MWh	Electricity consumed from the grid	Continuous meter reading; monthly and annual aggregation	Import meter/invoices	Used for project emissions from electricity consumption
Q_biogas,y	Nm3 or m3	Biogas flow to engines/flare	Continuous flow measurement; aggregated monthly and annually	Biogas flow meters/SCADA	Meter calibration and data completeness checks

Data / Parameter	Unit	Description	Frequency	Monitoring equipment/source	QA/QC procedures
CH4_biogas,y	% v/v	Methane content of biogas	Continuous analyzer or periodic measurement per monitoring plan	Gas analyzer/lab records	QA/QC and calibration required
Flare operation	hours/flow/temperature	Emergency flare operation and destruction conditions	Event-based and continuous where applicable	Flare logs/SCADA	Demonstrate combustion and destruction efficiency
Fossil fuel use	litres, kg, or GJ	Fossil fuels consumed by project equipment	As consumed; monthly/annual aggregation	Fuel purchase/use records	Apply TOOL03 where applicable
RDF production/end use	tonnes	RDF produced and sent to end users or otherwise managed	Operational records; monthly/annual aggregation	RDF production/offtake records	Needed for RDF/SB leakage/end-use assessment
Residual waste/by-products	tonnes	Residuals, digestate, leachate or by-products sent to disposal/treatment/use	Operational records; monthly/annual aggregation	Facility logs/manifests	Needed for leakage and project emissions

6.3 Monitoring Plan

Monitoring will be implemented by trained facility personnel under the responsibility of the project proponent/facility manager. Data will be collected from calibrated meters, SCADA/plant records, weighbridge records, waste characterization studies, invoices, laboratory analyses, operating logs, and maintenance/calibration records.

- *Waste reception data will be recorded through weighbridge or equivalent systems and reconciled with municipal/operational records.*
- *Electricity exported to and imported from the grid will be recorded using revenue-grade meters and reconciled with grid operator or settlement records.*
- *Biogas flow, methane content, engine operation, and flare events will be recorded through meters, analyzers, SCADA, and operating logs.*
- *RDF production, residual waste, digestate, leachate, and by-product movements will be recorded using facility logs, weighbridge records, manifests, and off-take/disposal documents.*
- *Monitoring equipment will be calibrated according to manufacturer recommendations, applicable standards, and the validated monitoring plan.*
- *Data will be archived electronically and/or physically for the period required by the VCS Program and made available to the VVB during verification.*
- *Missing data, meter failures, non-conformances, and corrective actions will be documented using conservative procedures consistent with the methodology and monitoring plan.*

7 APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
None identified in this draft	No commercially sensitive information has been intentionally excluded.	N/A

8 APPENDIX 2: DATA GAPS AND ASSUMPTIONS REGISTER

Topic	Gap / assumption	Needed evidence
Emission reductions	Annual baseline, project, leakage, and net ER values are not provided in the intake.	Validated emission reduction calculation spreadsheet.
Project scale	Intake contains a scale class and a source scale label that may not be aligned.	Final ER estimates and VCS scale classification check.
Stakeholder consultation	No project-specific consultation date, attendance, comments, or outcomes provided.	Consultation plan, notices, attendance, minutes, comments, and responses.
Permits and legal compliance	EMRA license is referenced, but full permit set is not included.	Generation license, environmental permits, waste management authorizations, grid connection evidence, EIA/EIA exemption if applicable.
Additionality	Investment/common practice inputs and conclusion are not provided.	TOOL02/TOOL27/TOOL24 analysis, inputs, sources, and sensitivity tests.
Waste characterization	Biomethanization-suitable fraction is provided, but detailed composition support is not included.	Waste characterization study and sampling procedure.
Monitoring equipment	Meter/analyzer model, serial number, accuracy, and calibration details are not provided.	Equipment inventory and calibration certificates.